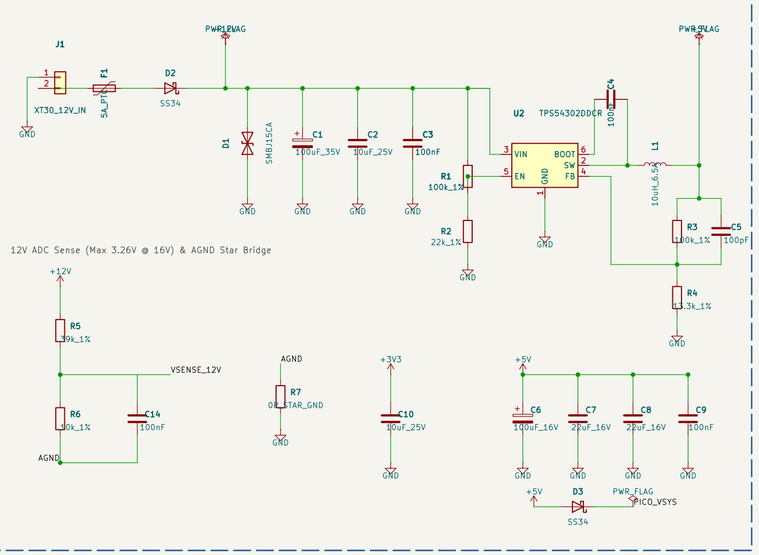
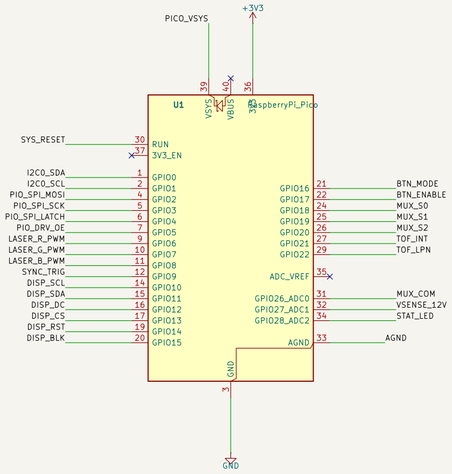
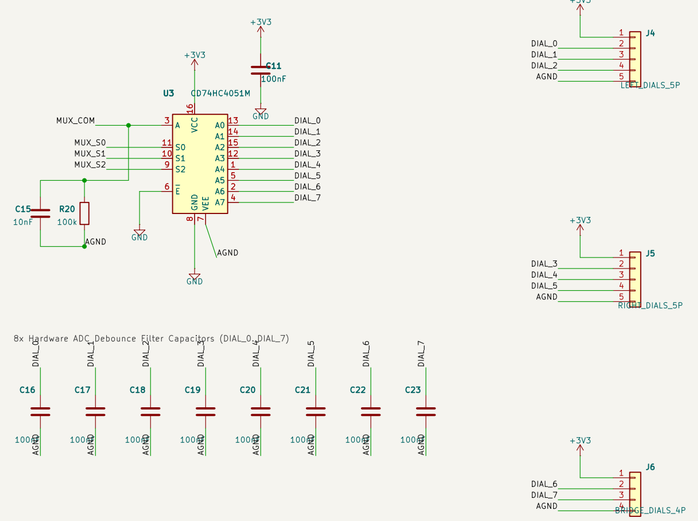
1. 12V POWER INPUT, PROTECTION & TPS54302 5V/3A BUCK STAGE



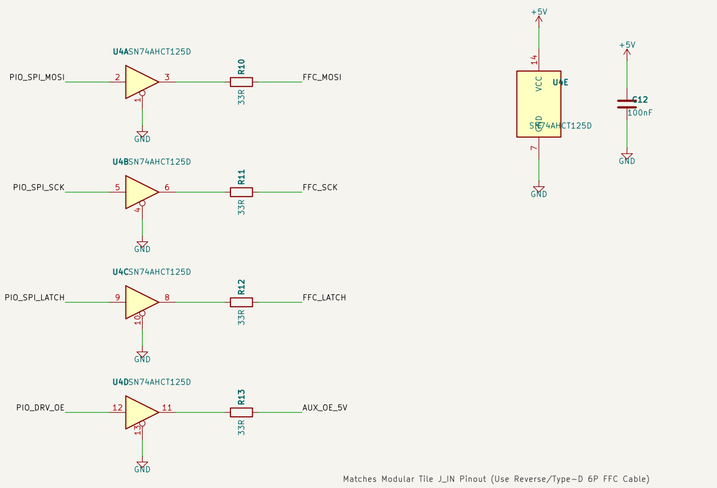
2. RASPBERRY PI PICO MASTER DOCK (RP2040 / RP2350)



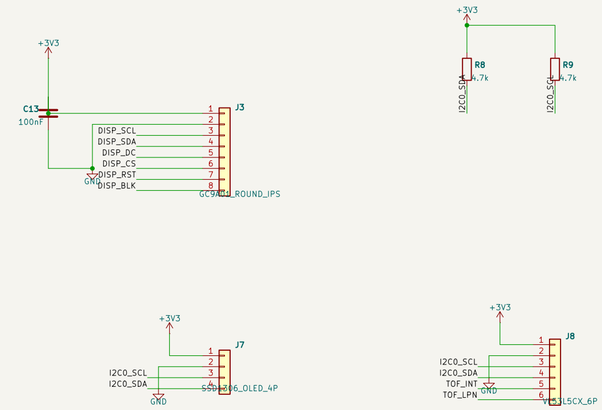
3. 74HC4051 8-CHANNEL ANALOG DIAL MUX & KNOB PORTS



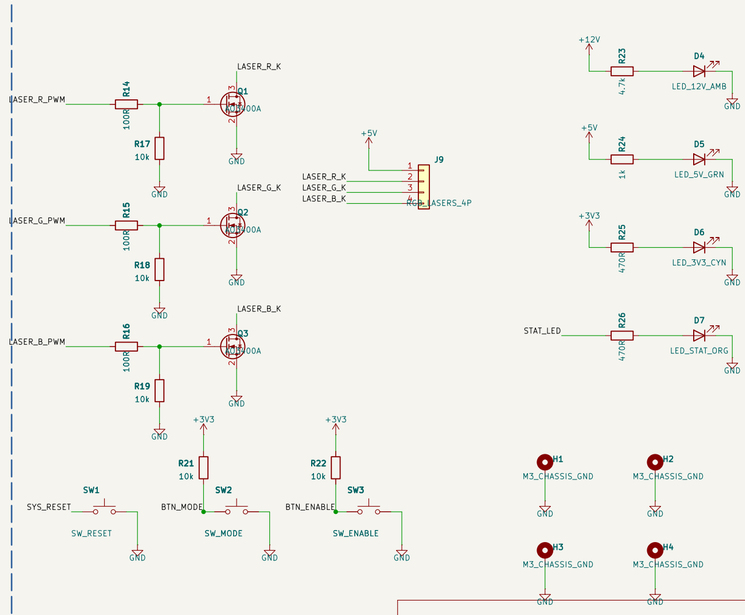
4. 5V SPI LEVEL SHIFTER, 33R DAMPING & HIROSE 6P FFC LINK



5. GC9A01 ROUND DISPLAY, SSD1306 OLED HUD & VL53L5CX TOF



6. AO3400A RGB LASER DRIVERS, CONTROLS, LEDS & M3 CHASSIS



HPVE Phased Array Volumetric Projection Engine – Production Schematic
TPS54302 5V/3A Buck + RP2040 Pico + CD74HC4051 8-Dial MUX + SN74AHCT125 5V SPI + GC9A01

Iron Cove Industries

Sheet: /
File: HPVE_Master_Carrier.kicad_sch

Title: HPVE Master Carrier Avionics Motherboard (MK-I)

Size: A2	Date: 2026-11-12	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 1/1